

4. Transport :-

layer 4 of OSI model plays a vital part in transmitting data across a network and little bit difficult to grasp.

* When data is sent between devices, it follows one of two different protocols that are decided based upon several factors

i) TCP ii) UDP

i) TCP :- Transmission Control protocol. potentially hinted by name, this process is designed with reliability and guarantee in mind. These protocol receives constant connection b/w two devices for amount of time it takes for data to be sent and received.

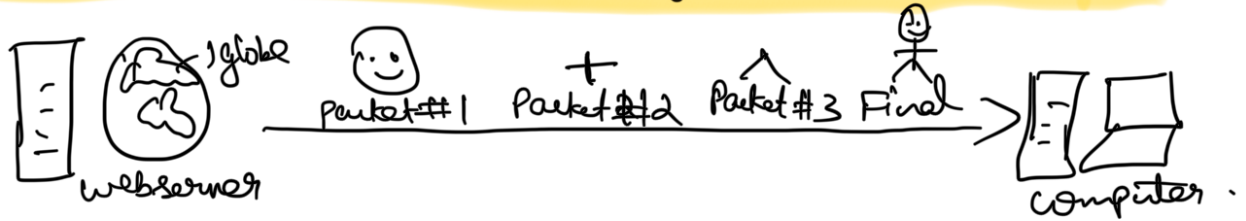
* TCP incorporates error checking into its design - Error checking is how TCP can guarantee that data sent from small chunks in session layer (layer 5) has then been received and reassembled in same order.

Advantages TCP	Disadvantage of TCP
Guarantees the accuracy of data	Requires reliable connection b/w 2 devices. If one small chunk of data is not received, then entire chunk of data cannot be received.
Capable of Synchronising two devices to prevent each other from being flooded with data.	A Slow connection can bottleneck another device as connection will be reserved on receiving computer whole time
performs lot more processes for reliability	TCP is significantly slower than UDP. because more work has to be done by devices using this protocol.

* TCP is used for situation such as file sharing, Internet browsing or sending an email. This usage is because these services requires data to be accurate and complete (no good having half a file).

In diagram below, we can see how a man is broken down into small pieces of data (known as packets) known as "webcamer", "stone

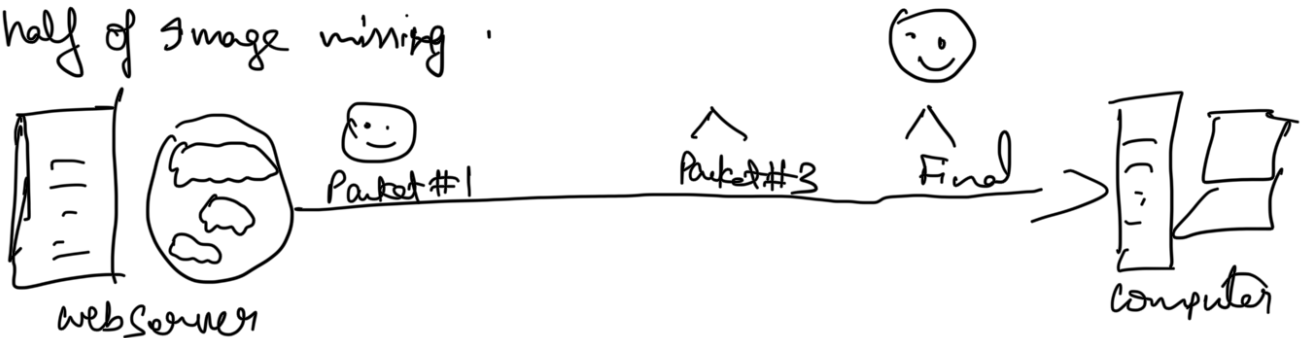
"Computer" reconstructs the picture of man into correct order.



UDP (User Datagram protocol) : — Tool is not advanced as TCP. It doesn't boast many features offered by TCP, such as error checking and reliability. There is no synchronization b/w two devices or guarantee.

Advantages of UDP	Disadvantages of UDP
* UDP is much faster than TCP	* UDP doesn't care if data is received
* UDP leaves application layer user software to decide if there any control over how quickly packets are sent	* It is quite flexible to software developers in this sense.
* UDP doesn't reserve a continuous connection on device TCP does.	* This means that unstable connections result in terrible experience for the user

* Using sample example as before, we can now see that only packets #1 and packets #3 have been received by "computer", meaning half of image missing.



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UDP is useful in situations where there are small pieces of data being sent. Ex protocols used for discovering devices (ARP and DHCP) or larger files such as video streaming (where it is okay if some portion of video is distorted).

fixed are just lost piece of data).

Quiz:-

- | | |
|--|---|
| 1) What is the name of this layer? | 2) What does TCP stand for? |
| A) Transport | A) Transmission control protocol |
| 3) What does UDP stand for? | 4) What protocol guarantees the accuracy of data? |
| A) User data gram protocol | A) TCP |
| 5) What protocol does it care if data is received or not by other devices? | 6) What protocol would an application such as email client use? |
| A) UDP | A) TCP |
| 7) What protocol would an application that downloads files use? | 8) What protocol would an application that streams videos use? |
| A) TCP | A) UDP |

5. Session layer:-

Once data has been correctly translated or formatted from the presentation layer (layer 6), the session layer (layer 5) will begin to create and maintain connection to other computer for which data is destined. When connection is established, a session is created. While connection is active, so is the session.

The session layer is responsible for closing the session if it hasn't been used for while or it is lost. Additionally a session can contain "checkpoints", if data is lost, only newest piece of data required to be sent, saving bandwidth.

What is why is Sessions are unique - meaning data cannot travel over different sessions.

Quiz:-

1) what is the name of this layer?

A) Session

2) what is the technical term for when a connection is successfully established?

A) Session